Particles and Nuclei 2

- 2/26/26
- Announcements:

- HW 3 Due Tuesday

- (Write down?)
- Reading: (Griff Ch 6) – Equivalent material in Mann (Free textbook) Ch 9 & 10: (~can skip 9.3, and more skim Ch 10, ABB toy theory
  - *Note Mann uses old time goes upward convention for Feynman diagrams*
  - *Includes discussion of Breit Wigner for resonances !!!*

Ch 5 – Only Few Quick Points

- **Quarkonia:**
- Heavy Quark Theory Generally Easier
  - (Also another advantage of looking for Tetra/Penta's with these)
  - Non-perturbative effects small: quark Mass itself already provides enough energy for Perturbation Theory

- Modern Theoretical Tools for Heavy Quark Physics:
  - NRQCD – (NonRelativ.) like in Griff – Shro. Equation (non-
  - pNRQCD – (Potential) –include various potentials mimicking Non-Pert. light quark
  - HQET SCET (soft collinear) Effective Theories including formation (maybe decay)
- Still nowadays Lattice QCD will be added to these
  - E.g. generate potentials from Lattice QCD for pNRQCD

- **"Separation of Scales"** all EFT's Even pQCD:
  - "Quark/Glue Structure Functions" (soft scale) convoluted w/ (hard scale) pQCD cross sections we are about to learn about

Ch. 5 Color/Flavor/Spin

- Just to make a few points about Ch 5
- So one main point is to "Derive" the (SOME) properties of the hadrons from the "quark" content
- E.g. enforcing symm/antisymm in WF
  - WF Asymmetry

$$\psi(total) = \psi(space) \cdot \psi(spin) \cdot \psi(flavour) \cdot \psi(colour)$$

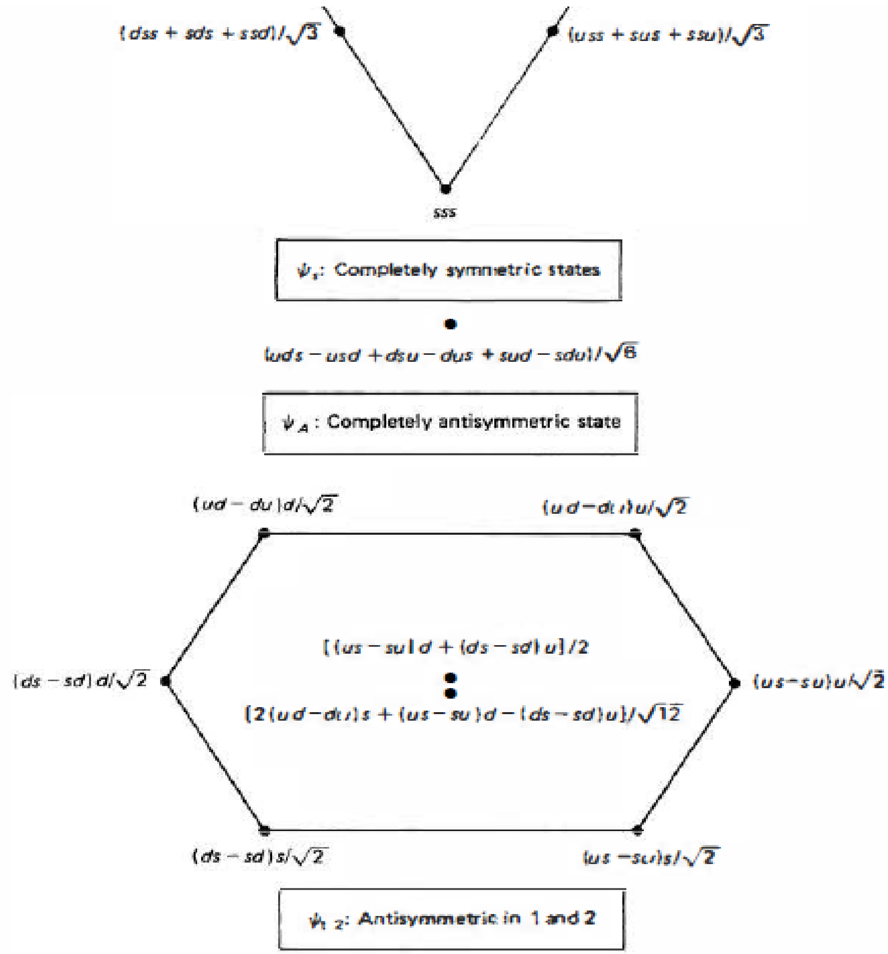
- Interesting: **<u>Baryons:</u>** common explanation (each q has one color) not accurate!
- In fact all quarks in one hadron have mixture of all three colors!!!

<u>All free naturally-occuring particles belong to a colour singlet</u>

$$\psi(colour) = \frac{1}{\sqrt{6}}(rgb - rbg + gbr - grb + brg - bgr)$$

Chapter 5

- The Wavefunctions -- OK now at the quark level…
- This trumps isospin
  - which wasn't perfect sym anyway.
  - quark flavors are not a SU(6) symmetry
- However, the "bashing" continues :
- These wavefunction combos are like that for which we are already experts at : SPIN



Flavor of p/n : not simple uud/ddu:

Ch. 5 Redux : Flavor/Spin

- Just to make a few more points about Ch 5 particularly
- So one main point is to "Derive" the properties of the hadrons from the "quark" content
- E.g. enforcing symm/antisymm in WF
  - WF Asymmetry

$$\psi(total) = \psi(space) \cdot \psi(spin) \cdot \psi(flavour) \cdot \psi(colour)$$

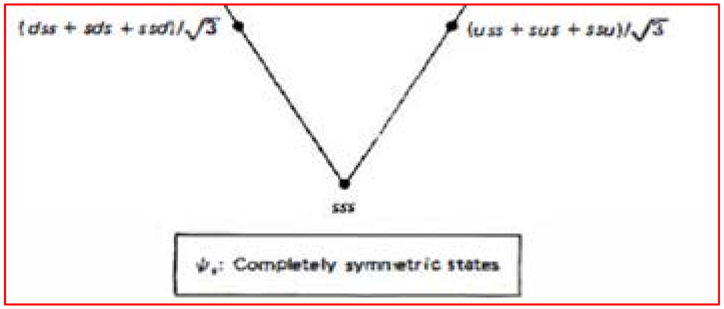
- Limited usefulness: No big driver for enumerating states (interest-wise) -- a small Jlab community (PWA)

$$3 \otimes 3 \otimes 3 = 10 \oplus 8 \oplus 8 \oplus 1$$

Symm Mixed Sym (12,23 or 13) AntiSymm

Example Omega Minus

- Flavor "sss" or symmetric (part of "fully symmetric decouplet



- Spin 3/2 : only symmetric

- Can't have mixed spin symmetries (needed for spin ½)

Look at Ω^- (sss) in the PDG.

There is $\Omega(1672)^-$ $J^P = 3/2^+$

$\Omega(2250)^-$ $J^P =$?? presumably $(\ell \text{ or } \ell' = 0)$

No spin 1/2 version (which would presumably be lighter than the spin 3/2 version).

- Similarly, often when constructing such enumerations we find " **missing states** "

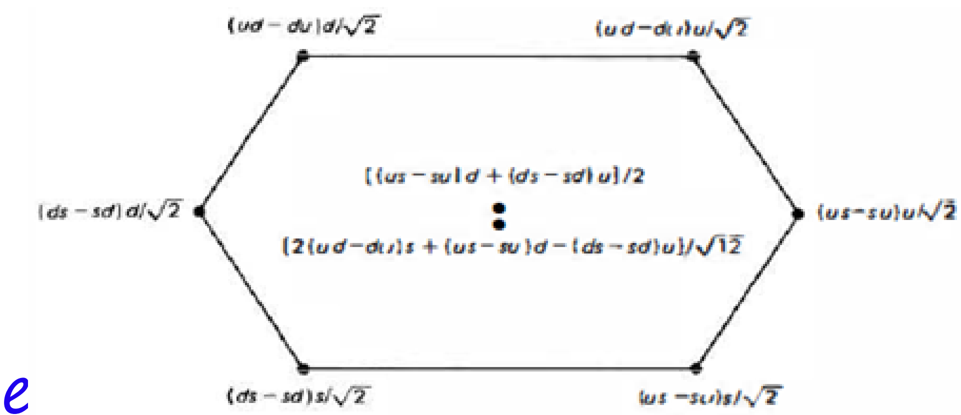
Proton/Neutron

- Mixed Symmety case:

For the total wavefunction, the (mixed symmetry) flavour octet members must be have be matched to the $\psi(spin)$ having the corresponding (anti-)symmetry:

$$\psi(baryon\ octet) =$$

$$\frac{\sqrt{2}}{3}\left[\psi_{12}(spin)\psi_{12}(flavour) + \psi_{23}(spin)\psi_{23}(flavour) + \psi_{13}(spin)\psi_{13}(flavour)\right]$$



Flavor of p/n : not simple uud/ddu:

Another Example Meson

- Another reason rho to 2pi0 not allowed

Consider the decay $\rho^0 \to \pi\pi$. At first glance, it would seem that one would expect two final states (for decays into two pions – which is the case almost 100% of the time):

$$\rho^0 \to \pi^+\pi^- \qquad \rho^0 \to \pi^0\pi^0$$

Can only be symmetric
(color aysmmetric)
rho WF anti

Ch. 5 Redux : Flavor/Spin

- Ch 5
- So one main point is to "Derive" the properties of the hadrons from the "quark" content
- E.g. enforcing symm/antisymm in WF
  - WF Asymmetry

$$\psi(total) = \psi(space) \cdot \psi(spin) \cdot \psi(flavour) \cdot \psi(colour)$$

- But point made before: These considerations only apply to "Valence quarks" or "dressed quarks"
- Because we know the hadron structure is must more complicated – NON-PERTURBATIVE → there must be much more to it than these simplistic considerations
- REALLY NEED LATTICE QCD! (or full Non Perturb)

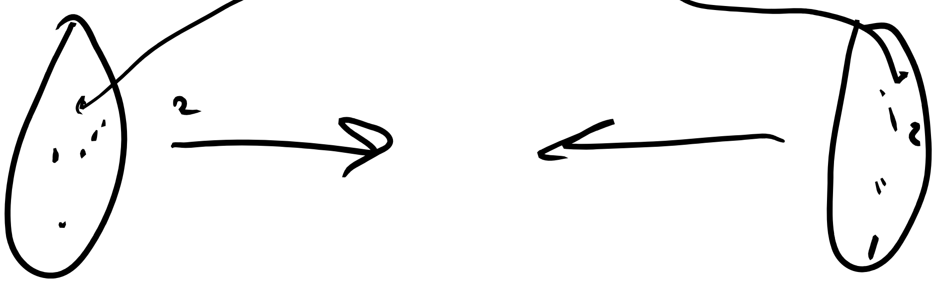
Parton Structure Functions– Better Description

- "Factorization" Separation of Scales

- F(x) → "PDF" Parton Distrib. Function:

$$\boxed{\sigma \sim \iiint F(x_1)\ \ F(x_2)\ \ \hat{\sigma}\ \ D(z_1)\ \ D(z_2).}$$

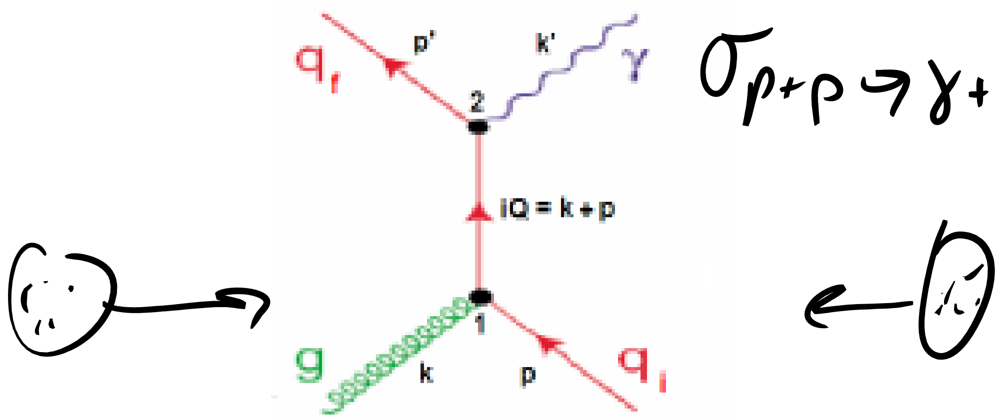
$x \equiv$ parton Fraction of hadron mom



- D(z)'s → Outgoing Formation from quarks to hadrons

Example

- Justin's Thesis: Direct Photons Compton Scattering qg→qγ:



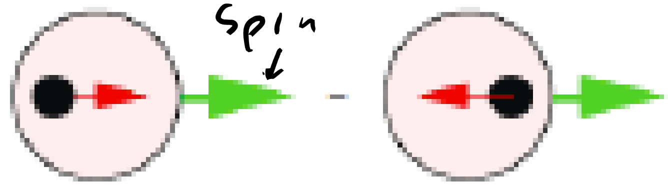
$\sigma_{p+p\rightarrow\gamma+X}$ parton dist fns

$G^{a/N}$

$$\frac{d\sigma_{pp\rightarrow C+X}}{d^3p_C} = \int dx_{q_i} dx_g dz_q G_{q_i/p}(x_{q_i}, M_\gamma^2) G_{g/p}(x_g, \mu_F^2 a) D_{C/c}(z_C, \mu_F a^2) \times \frac{\hat{s}}{z_C^2 \pi} \frac{d\sigma_{qg\rightarrow qc}}{d\hat{t}} \delta(\hat{s}+\hat{t}+\hat{u})$$

Spin-Dependent PDF's

- Using polarized protons

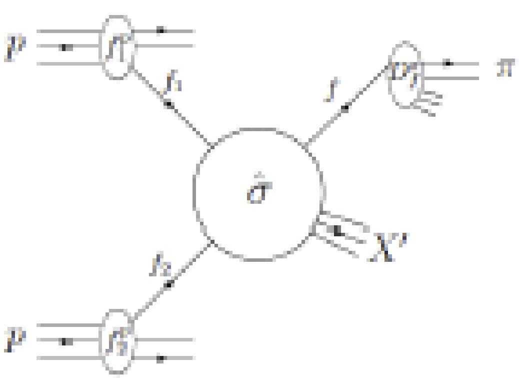


spin

fraction hadron spin

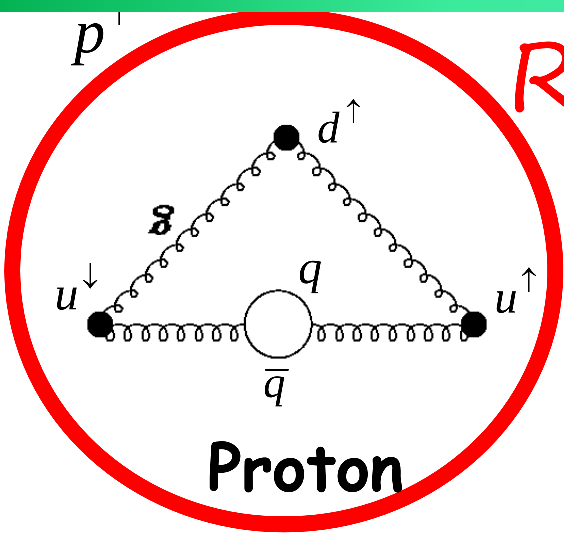
x_i

$$\Delta f(x,\mu) \equiv f_{\rightarrow}^{\rightarrow}(x,\mu) - f_{\leftarrow}^{\rightarrow}(x,\mu)$$



$$\frac{d\Delta\sigma^{pp\to\pi X}}{dP} = \sum_{f_1,f_2,f}\int dx_1 dx_2 dz\, \boxed{\Delta f_1^p(x_1,\mu^2)\, \Delta f_2^p(x_2,\mu^2)} \times \frac{d\Delta\hat{\sigma}^{f_1 f_2\to f X'}}{dP}(x_1,p_1,x_2,p_2,p_\pi/z,\mu)\, D_f^\pi(z,\mu^2)\,,$$

The Proton Spin Crisis



REMINDER

Say you have a proton with total spin +1/2 along some axis. You'd expect it to contain two quarks with spin +1/2 and one with spin -1/2.
1/2 + 1/2 - 1/2 = +1/2

Surprising data from polarized muon-nucleon scattering in late 1980s!
1987: Only 12% +- 16% of proton's spin carried by quarks!

The proton spin crisis begins!!

Quark Spin Gluon Spin

Spin $$\frac{1}{2} = \frac{1}{2}\Delta\Sigma + \Delta G + L_z$$

Orbital Angular Momentum

The rest now expected to be from gluon spin and orbital angular momentum of quarks and gluons, but this hasn't been easy to measure!

Griffiths Ch 5 model: Complete BS?

- It seems theses model of Griffiths does quite well. What to make of that?
- GRIFF BOOK ONLY : Lattice QCD calcs don't consider anything about Spin-Spin Interactions
- Notice "fitting" ~very few states -- Not many states out of the literally ~hundred(s?). (1 to 3 parameter -- 7 states)

Table 5.3 Pseudoscalar and vector meson masses. (MeV/c^2)

Meson	Calculated	Observed
π	139	138
K	487	496
η	561	548
ρ	775	776
ω	775	783
K^*	892	894
ϕ	1031	1020

in place of a u or d. But, that cannot be the whole story, f
weigh the same as the π's; after all, they have the same qu
the spatial ground state ($n = 1, l = 0$). Since the pseudosca
fer only in the relative orientation of the quark spins, the
must be attributed to a spin–spin interaction, the QCD ar
in the ground state of hydrogen. This suggests the follow

$$M(\text{meson}) = m_1 + m_2 + A\frac{(\mathbf{S}_1 \cdot \mathbf{S}_2)}{m_1 m_2}$$

$$M(\text{baryon}) = m_1 + m_2 + m_3 + A'\left[\frac{\mathbf{S}_1 \cdot \mathbf{S}_2}{m_1 m_2} + \frac{\mathbf{S}_1 \cdot \mathbf{S}_3}{m_1 m_3} + \frac{\mathbf{S}_2 \cdot \mathbf{S}_3}{m_2 m_3}\right]$$

Table 5.6 Baryon octet and decupiet masses. (MeV/c^2)

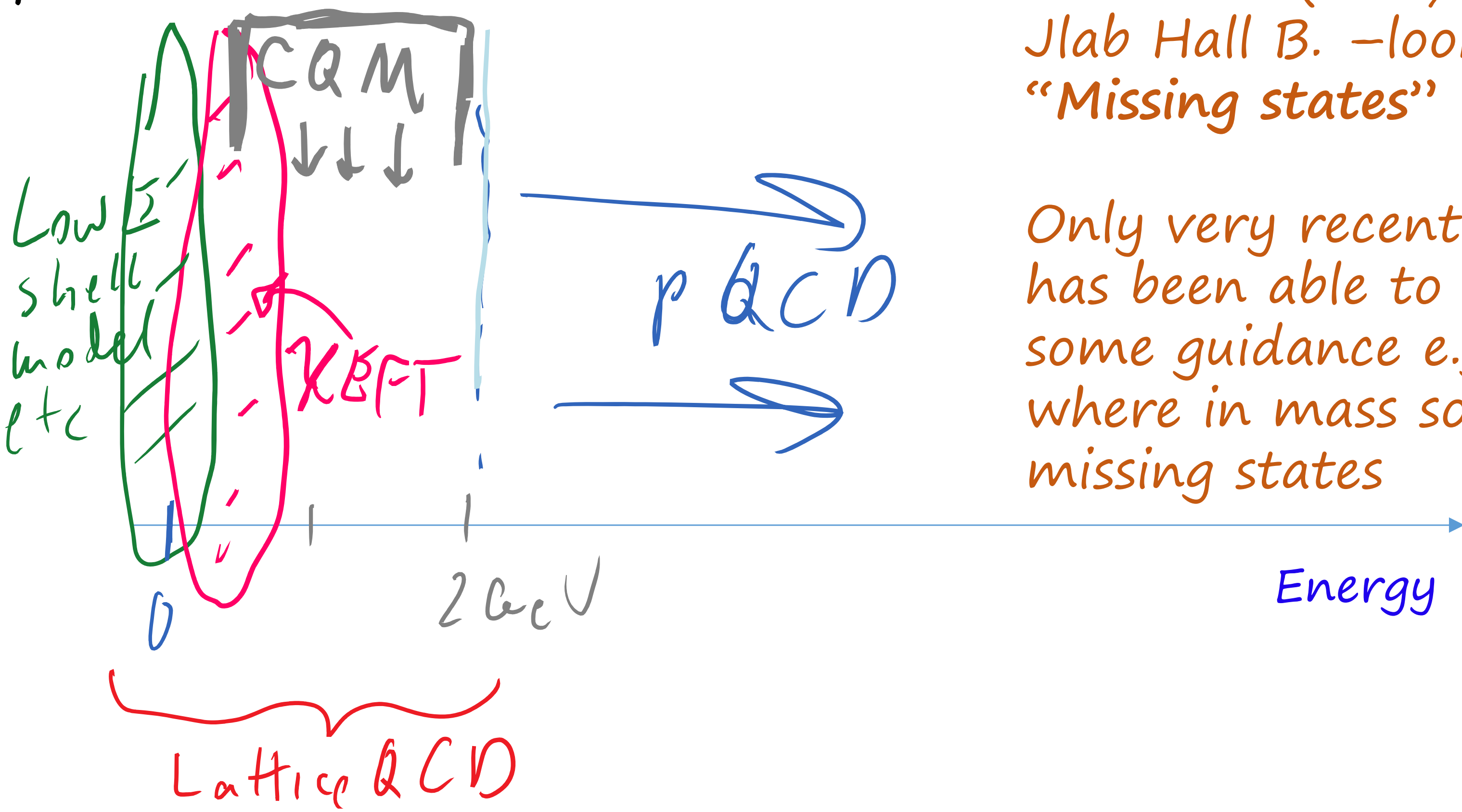
Baryon	Calculated	Observed
N	939	939
Λ	1114	1116
Σ	1179	1193
Ξ	1327	1318
Δ	1239	1232
Σ^*	1381	1385
Ξ^*	1529	1533
Ω	1682	1672

Summary / Interesting- Pions

- Another strike against Constiutuent Quark Model (CQM): Pion lightest meson!): no "constituent" quarks? Chi EFT – Debated – See next slides .
- Griffiths idea perhaps is to make us feel better about filling the "gap" between perturbative QCD and non-perturbative/lowE

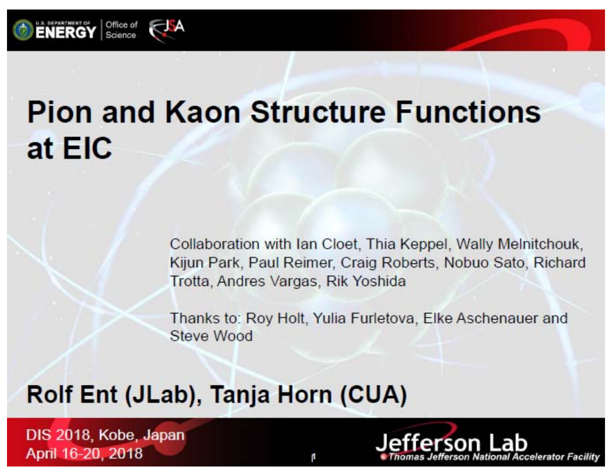
Ken Hicks (Jlab) CLAS Jlab Hall B. –looking for "**Missing states**"

Only very recently LQCD has been able to provide some guidance e.g. as to where in mass some of the missing states



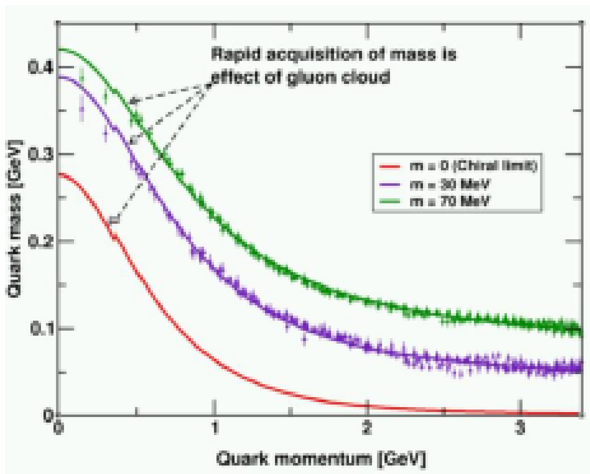
Strong feelings: Pion Structure

- stackexchange – strong opinions about pion being "goldstone" boson of QCD, which is something more like a "phonon" pseudostate/particle –technically made of QCD particles, ie quarks/glue, but not in a way well described by the usual CQM – e.g. larger scale coherence of the strong fields themselves.
  - **Lattice QCD shows for example, that if the small (~3 MeV) quark Higgs masses are set to 0, the proton will retain its full 1 GeV/300 MevCQ mass (dynamically generated by fields/gluons making the dressed quarks), while the pion <u>will become completely massless!</u>**
- This is an active area of research---EIC: study the pQCD parameterization of the gluon "structure functions" inside the pions!



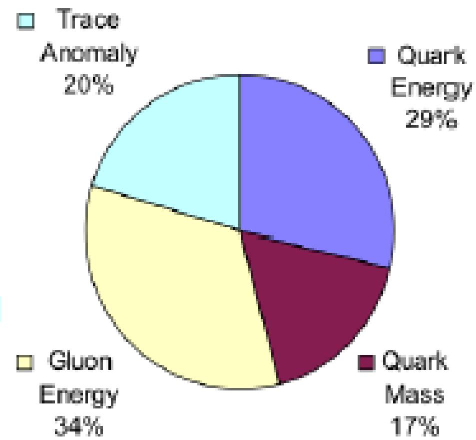
The Incomplete Nucleon: Mass Puzzle

"… The vast majority of the nucleon's mass is due to quantum fluctuations of quark-antiquark pairs, the gluons, and the energy associated with quarks moving around at close to the speed of light. …"



"Mass without mass!"

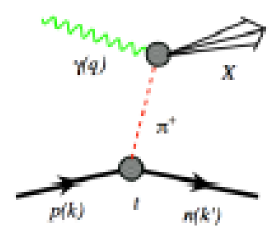
Bhagwat & Tandy/Roberts et al



- **EIC's expected contribution in:**
  - Quark-gluon energy:
    $\propto$ *quark-gluon momentum fractions*

In nucleon with DIS and SIDIS

In pions and kaons with Sullivan process



Proton: Mass ~ 940 MeV
preliminary LQCD results on mass budget, or view as mass acquisition by DχSB

Kaon: Mass ~ 490 MeV
at a given scale, less gluons than in pion

Pion: Mass ~ 140 MeV
mass enigma – gluons vs Goldstone boson

Origin of mass of QCD's pseudoscalar Goldstone modes

- Exact statements from QCD in terms of current quark masses due to PCAC
  (Phys. Rep. 87 (1982) 77;
  Phys. Rev. C 56 (1997) 3369;
  Phys. Lett. B420 (1998) 267)

$$f_\pi m_\pi^2 = (m_u^\zeta + m_d^\zeta)\rho_\pi^\zeta$$
$$f_K m_K^2 = (m_u^\zeta + m_s^\zeta)\rho_K^\zeta$$

- Pseudoscalar masses are generated dynamically – If $\rho_\pi \neq 0$, $m_\pi^2 \sim \sqrt{m_q}$
  - The mass of bound states increases as $\sqrt{m}$ with the mass of the constituents
  - In contrast, in quantum mechanical models, *e.g.*, constituent quark models, the mass of bound states rises linearly with the mass of the constituents
  - *E.g.*, with constituent quarks Q: in the nucleon $m_Q \sim \frac{1}{3} m_N \sim 310$ MeV, in the pion $m_Q \sim \frac{1}{2} m_\pi \sim 70$ MeV, in the kaon (with one s quark) $m_Q \sim 200$ MeV – **This is not real**.
  - In both DSE and LQCD, the mass function of quarks is the same, regardless what hadron the quarks reside in – **This is real**. It is the Dynamical Chiral Symmetry Breaking (DχSB) that makes the pion and kaon masses light.
- Assume DχSB similar for light particles: If $f_\pi = f_K \approx 0.1$ & $\rho_\pi = \rho_K \approx (0.5\ \text{GeV})^2$ @ scale $\zeta = 2$ GeV
  - $m_\pi^2 = 2.5 \times (m_u^\zeta + m_d^\zeta)$; $m_K^2 = 2.5 \times (m_u^\zeta + m_s^\zeta)$
  - Experimental evidence: mass splitting between the current s and d quark masses

$$m_K^2 - m_\pi^2 = (m_s^\zeta - m_d^\zeta)\frac{\rho^\zeta}{f} = 0.225\,\text{GeV}^2 = (0.474\,\text{GeV})^2 \qquad m_s^\zeta = 0.095\,\text{GeV},\ m_d^\zeta = 0.005\,\text{GeV}$$

In good agreement with experimental values

The role of gluons in pions

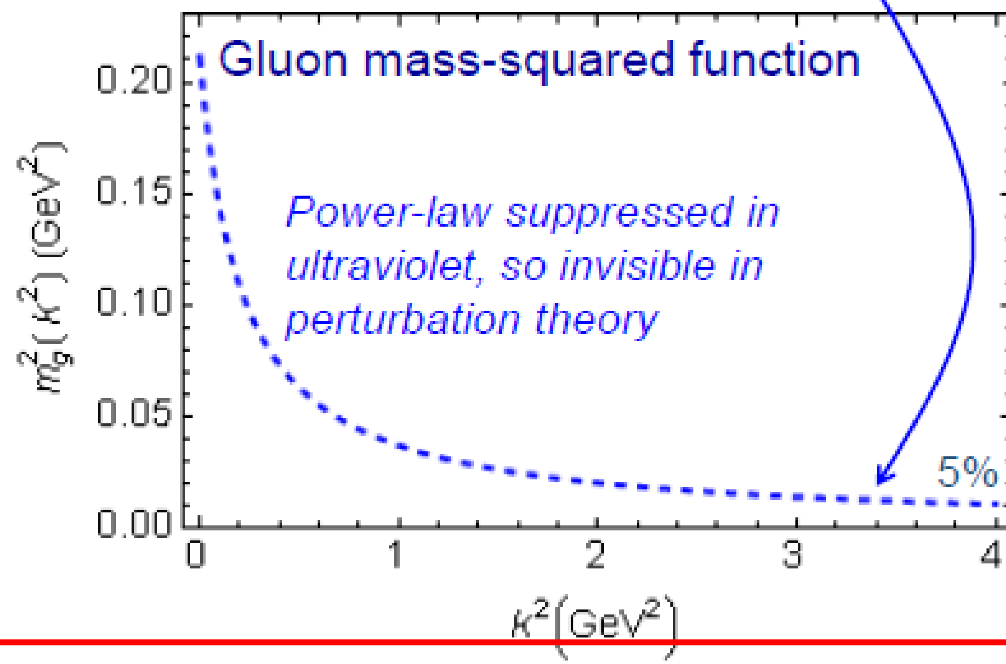
Pion mass is enigma – cannibalistic gluons vs massless Goldstone bosons

$$f_\pi E_\pi(p^2) = B(p^2)$$

$$m_g^2(k^2) = \frac{\mu_g^4}{\mu_g^2 + k^2}$$

Adapted from Craig Roberts:

- The most fundamental expression of Goldstone's Theorem and DCSB in the SM
- Pion exists if, and only if, mass is dynamically generated
- This is why $m_\pi = 0$ in the absence of a Higgs mechanism



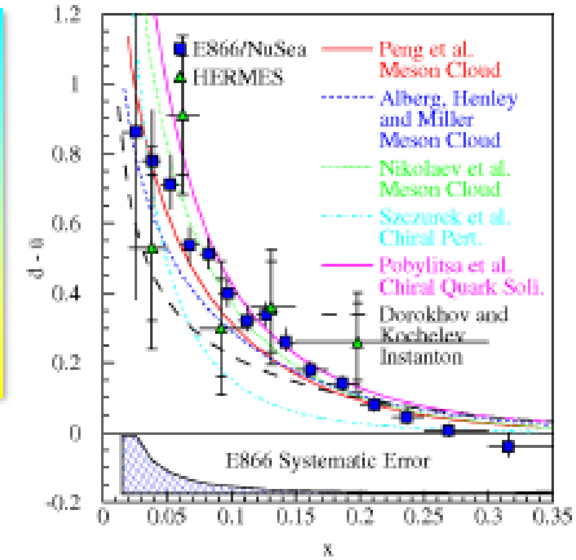
What is the impact of this for gluon parton distributions in pions vs nucleons?

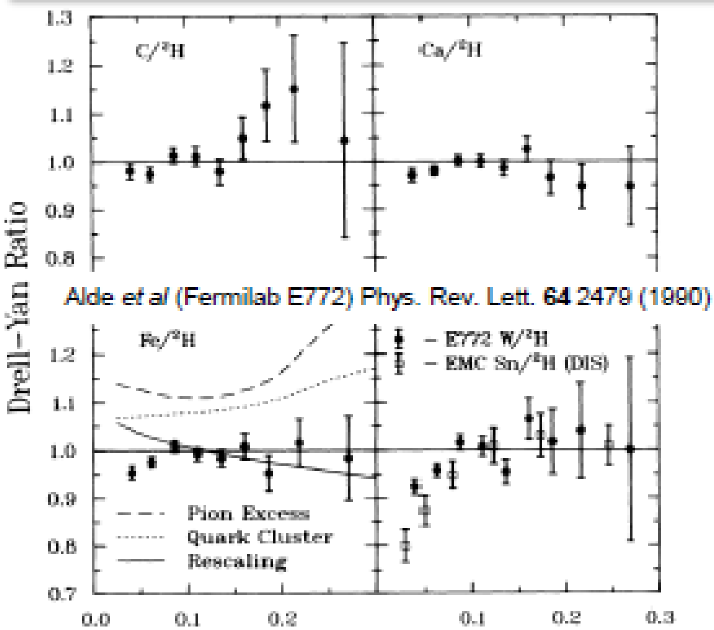
One would anticipate a different mass budget for the pion and the proton

Why should you be interested in pions and kaons?

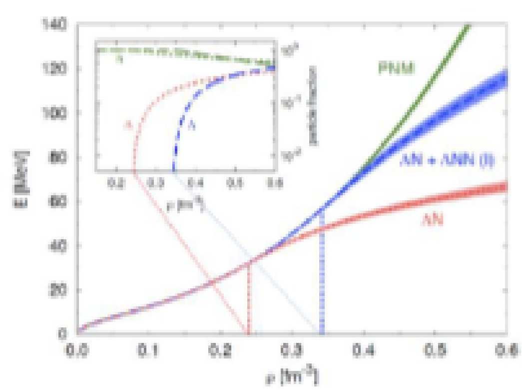
Protons, neutrons, pions and kaons are the main building blocks of nuclear matter

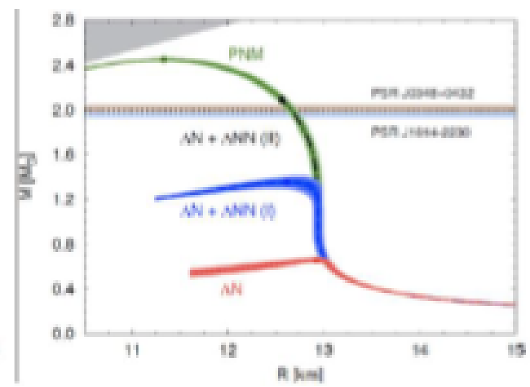
1) The pion, or a meson cloud, explains light-quark asymmetry in the nucleon sea
2) Pions are the Yukawa particles of the nuclear force – but no evidence for excess of nuclear pions or anti-quarks
3) Kaon exchange is similarly related to the ΛN interaction – correlated with the Equation of State and astrophysical observations
4) Mass is enigma – cannibalistic gluons vs massless Goldstone bosons





Alde *et al* (Fermilab E772) Phys. Rev. Lett. **64** 2479 (1990)





Chapter 6

- We already discussed some of the basics of the REAL differential cross section and rates/decay distributions
- (Invariant)

$$\cancel{\frac{d\sigma}{d\Omega}} = \frac{d\sigma}{d\Omega_p} = \frac{dn}{d\Omega_p} \propto \frac{dn}{d\theta_p \, d\phi_p}\left(\frac{1}{\sin\theta_p}\right)$$

$$E\frac{dn}{d\vec{p}^{\,3}} = \frac{dn}{dp_x dp_y dp_z}$$

$$d\vec{p}^{\,3} = dp_x \, dp_y \, dp_z$$

$$= p^2 d\Omega_p = p^2 \sin\theta_p \, d\theta_p \, d\phi_p \, d\vec{p}$$

$$= p_\perp dp_\perp \, dp_z \, d\phi_p$$

$$\frac{dn_{decay}}{d\theta_\gamma d\phi_\gamma} \Rightarrow \frac{dn_{decay}}{d\vec{p}^{\,3}_\gamma / E}$$

Golden Rule Review

transition probability $W = \frac{2\pi}{\hbar} \left| \mathcal{M}_{if} \right|^2 \rho_f(E)$ Density of final states available for energy E.

matrix element $\left| \mathcal{M}_{if} \right|^2 = \left| \langle \psi_f | V_{if} | \psi_i \rangle \right|^2$ Matrix element contains the fundamental physics (e.g. the dynamics).

- Transform these to two specific situations
- Make two Golden Rule Expressions from further details of imposing Lorentz Invariance / Field Theory considerations

Griff Notation Won't be Used

- Griff formula: d4p COMPLETELY NON STANDARD

Suppose particle 1 (at rest)[†] decays into several other particles 2, 3, 4, .. ., n:

$$1 \to 2 + 3 + 4 + \ldots + n \tag{6.14}$$

The decay rate is given by the formula

$$\Gamma = \frac{S}{2\hbar m_1} \int |\mathscr{M}|^2 (2\pi)^4 \delta^4 \left(p_1 - p_2 - p_3 \cdots - p_n\right) \times \prod_{j=2}^{n} 2\pi\,\delta\left(p_j^2 - m_j^2 c^2\right) \theta\left(p_j^0\right) \frac{d^4 p_j}{(2\pi)^4} \tag{6.15}$$

$$\Gamma = \frac{S}{2\hbar m_1} \int |\mathscr{M}|^2 (2\pi)^4 \delta^4 \left(p_1 - p_2 - p_3 \cdots - p_n\right) \times \prod_{j=2}^{n} \frac{1}{2\sqrt{\mathbf{p}_j^2 + m_j^2 c^2}} \frac{d^3 \mathbf{p}_j}{(2\pi)^3} \tag{6.21}$$

Master Golden Rules

Golden Rule for Decays

Statistical Factor

Consider decays of the form 1➔ 2+3+ ……..+ N

$$d\Gamma = |\mathcal{M}|^2 \frac{S}{2\hbar m_1}\left[\left(\frac{c d^3\vec{p}_2}{(2\pi)^3 2E_2}\right)\left(\frac{c d^3\vec{p}_3}{(2\pi)^3 2E_3}\right)\cdots\cdots\left(\frac{c d^3\vec{p}_n}{(2\pi)^3 2E_n}\right)\right]\times(2\pi)^4\delta^4(p_1 - p_2 - \ldots\ldots - p_n)$$

- For scattering

For the process 1 + 2 ➔ 3 + 4 + …. + N

$$d\sigma = |\mathcal{M}|^2 \frac{\hbar^2 S}{4\sqrt{(p_1\cdot p_2)^2 - (m_1 m_2 c^2)^2}}\left[\left(\frac{c d^3\vec{p}_3}{(2\pi)^3 2E_3}\right)\left(\frac{c d^3\vec{p}_4}{(2\pi)^3 2E_4}\right)\cdots\cdots\left(\frac{c d^3\vec{p}_n}{(2\pi)^3 2E_n}\right)\right]\times(2\pi)^4\delta^4(p_1 + p_2 - \ldots\ldots - p_n)$$

These are the forms used in Mann

Simplified versions: Our Master Formulae

- 1→ 2+3 Decay COM

more generally

$$\Gamma = \frac{S|\mathbf{p}|}{8\pi\hbar m_1^2 c}|\mathcal{M}|^2$$

$$\boxed{\frac{1}{\tau} = \Gamma = \frac{|\vec{p}^*|}{32\pi^2 m_i^2}\int |M_{fi}|^2 \mathrm{d}\Omega}$$

Griff's

Thomson

- Cross section 1+2→3+4

$$\frac{d\sigma}{d\Omega} = \left(\frac{\hbar c}{8\pi}\right)^2 \frac{S|\mathcal{M}|^2}{(E_1+E_2)^2}\frac{|\mathbf{p}_f|}{|\mathbf{p}_i|}$$

$$\boxed{\sigma = \frac{1}{64\pi^2 s}\frac{|\vec{p}_f^*|}{|\vec{p}_i^*|}\int |M_{fi}|^2 \mathrm{d}\Omega^*}$$

Better invariant version

$$\boxed{\frac{\mathrm{d}\sigma}{\mathrm{d}t} = \frac{1}{64\pi s|\vec{p}_i^*|^2}|M_{fi}|^2}$$

Piece by Piece

- Let's say a little more about where the parts of the formulae come from

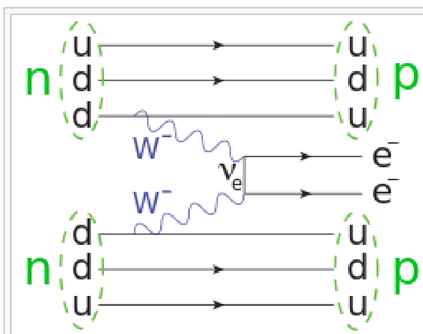
- E.g. 1→ 2+3 Decay COM

$$\Gamma = \frac{S|\mathbf{p}|}{8\pi\hbar m_1^2 c}|\mathcal{M}|^2$$

- **First:** statistical factor → A VERY QUANTUM PART!!!

where m_i is the mass of the ith particle and p_i is its four-momentum. S is a statistical factor that corrects for double-counting when there are identical particles in the final state: for each such group of s particles, S gets a factor of $(1/s!)$. For instance, if $a \to b + b + c + c + c$, then $S = (1/2!)(1/3!) = 1/12$. If there are *no* identical particles in the final state (the most common circumstance), then $S = 1$.

- $\pi^0 \to 2\gamma$ = ? **0νββ = ?**



2νββ = ?

Double beta decay
S=1/4= ½ * ½

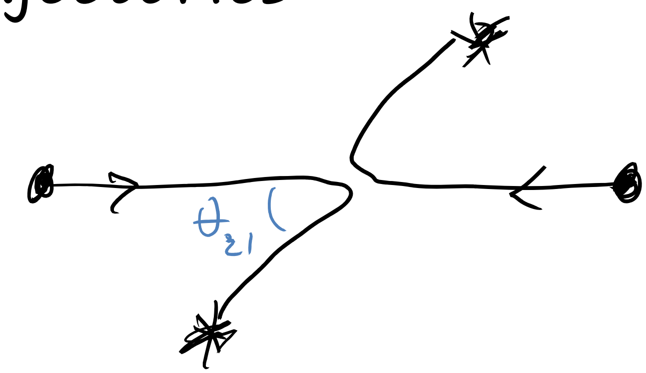
-

(from undergrad qm course)

Identical Particles

- All atoms identical Quantum Effect? (identicality)
  - No: Ancient Greeks had it in mind…
  - on other hand: Quantum Field Theory Assumes Each e.g. $e^{\pm}$ is quantum of same universal field
- Can't "paint red" == are not localized
- Quantum uncertainty in trajectories

Case D "deflect"



Case B "backscatter"

$\sigma \propto$ scattering rate $\propto |f(\theta)|^2$

$f(\theta) \sim \psi^{scatt}$

$|f(\theta_{12})| = |f(\theta_{21})| \rightarrow f(\theta) = \frac{1}{2}[f(\theta_{12}) \pm f(\theta_{21})]$

$|f(\theta)|^2 = 2\,|f(\theta_{21})|^2 +$ <u>Interference (f^*f)</u>

Identical Particles Have
Smaller/Larger σ! p+n:n+n 1:2

Exchange Force

- Similar to Cross Section Idea : Interference terms -- Demonstration of 2nd View
- Calc $\langle (x_1 - x_2)^2 \rangle$
- Distinguishable $\psi(x_1, x_2) = \psi_a(x_1)\psi_b(x_2)$:
$$\langle (x_1 - x_2)^2 \rangle_d = \langle x^2 \rangle_a + \langle x^2 \rangle_b - 2\langle x \rangle_a \langle x \rangle_b.$$
- Indistinguishable
$$\psi(x_1, x_2) = \psi_a(x_1)\psi_b(x_2) \pm \psi_b(x_1)\psi_a(x_2):$$
$$\langle (x_1 - x_2)^2 \rangle_\pm = \langle (x_1 - x_2)^2 \rangle_d \mp 2|\langle x \rangle_{ab}|^2,$$
Extra term
- For λ_P = -1 ("fermion" if spin symmetric), decorrelation (Δx ↑)! = +1: Δx ↓ Bose-Einstein Correlation.
  - Like a FORCE! → "5th?", "Geometric"/Quantum Force"
- Similar effect in particle production in hadron collisions, emission of photons from astrophysical sources: Hanbury Brown Twist Effect on <$k_1 k_2$>

Simplified versions

- 1→ 2+3 Decay COM

$$\Gamma = \frac{S|\mathbf{p}|}{8\pi\hbar m_1^2 c}|\mathcal{M}|^2$$

more generally

$$\boxed{\frac{1}{\tau} = \Gamma = \frac{|\vec{p}^*|}{32\pi^2 m_i^2}\int |M_{fi}|^2 \mathrm{d}\Omega}$$

- Cross section 1+2→3+4

$$\frac{d\sigma}{d\Omega} = \left(\frac{\hbar c}{8\pi}\right)^2 \frac{S|\mathcal{M}|^2}{(E_1+E_2)^2}\frac{|\mathbf{p}_f|}{|\mathbf{p}_i|}$$

$$\boxed{\sigma = \frac{1}{64\pi^2 s}\frac{|\vec{p}_f^*|}{|\vec{p}_i^*|}\int |M_{fi}|^2 \mathrm{d}\Omega^*}$$

Better invariant version

$$\boxed{\frac{\mathrm{d}\sigma}{\mathrm{d}t} = \frac{1}{64\pi s|\vec{p}_i^*|^2}|M_{fi}|^2}$$

Exercise

- For a decay 1 → 2+ 2
- (meaning single particle mass m1 decays to 2 identical particles with m2)
- derive an expression for $|p|$, either final state momentum in the COM frame in terms of the masses of the two particles

Example

$$\frac{\Gamma(D^0 \to \pi^-\pi^+)}{\Gamma(D^0 \to K^-\pi^+)} = \frac{|V_{cd}|^2}{|V_{cs}|^2}\left(\frac{p_{\pi^-}}{p_{K^-}}\right) \approx \lambda^2 \left(\frac{922}{861}\right)$$

MeV

e.g. kinematic enhancement factor is 1.07 but dynamical suppression factor is λ^2 ~ .05.

$m_{D^0} = 1.85\,\mathrm{GeV}$

$m/2 = 0.925$

$\sqrt{.92^2 - 0.138^2}$

≈ 0.92

(for pi basically all m/2 goes to momentum)

How did we get simplified versions?

$$\delta^4(p) \equiv \delta^4(p^\mu) = \delta(p_0)\delta(p_x)\delta(p_y)\delta(p_z)$$

Example same massless(or same) particles for 2+3

Consider decays of the form 1➔ 2+3+ ....+ N

$$d\Gamma = |\mathcal{M}|^2 \frac{S}{2\hbar m_1}\left[\left(\frac{c\,d^3\vec{p}_2}{(2\pi)^3 2E_2}\right)\left(\frac{c\,d^3\vec{p}_3}{(2\pi)^3 2E_3}\right)\cdots\cdots\left(\frac{c\,d^3\vec{p}_n}{(2\pi)^3 2E_n}\right)\right]\times(2\pi)^4\delta^4(p_1 - p_2 - \ldots\ldots - p_n)$$

$$\Gamma = \frac{S}{\hbar m}\left(\frac{1}{4\pi}\right)^2 \frac{1}{2}\int \frac{|\mathcal{M}|^2}{|\vec{p}_2||\vec{p}_3|}\delta\left(mc - \underbrace{|\vec{p}_2| - |\vec{p}_3|}_{E_{2,3}}\right)\delta^3(-\vec{p}_2 - \vec{p}_3)\,d^3\vec{p}_2\,d^3\vec{p}_3$$

$E_2 = E_3$

1→ 2+3 Decay (massless 2,3)

$$\Gamma = \frac{S}{\hbar m}\left(\frac{1}{4\pi}\right)^2 \frac{1}{2}\int \frac{|\mathcal{M}|^2}{\underbrace{|\vec{p}_2|}_{E_2}\underbrace{|\vec{p}_3|}_{=E_3}}\delta\left(mc - \underbrace{|\vec{p}_2| - |\vec{p}_3|}_{E_{2,3}}\right)\delta^3\left(-\vec{p}_2 - \vec{p}_3\right) d^3\vec{p}_2\, \cancel{d^3\vec{p}_3}$$

$$\Gamma = \frac{S}{2(4\pi)^2 \hbar m}\int \frac{|\mathcal{M}|^2}{\cancel{|\vec{p}_2|^2}}\delta\left(mc - 2|\vec{p}_2|\right)\cancel{d^3\vec{p}_2}\; p_2^2\, d\,p_2 \qquad \mathcal{M} = \mathcal{M}(|\vec{p}_2|).$$

$$\delta(kx) = \frac{1}{k}\delta(x)$$

$d^3\vec{p}_2 = |\vec{p}_2|^2\, d|\vec{p}_2| \sin\theta\, d\theta\, d\phi$ Do angular integral $\int d\Omega = 4\pi$

$$\Gamma = \frac{S}{8\pi\hbar m}\int_0^\infty |\mathcal{M}|^2 \frac{1}{2}\delta\left(|\vec{p}_2| - \frac{mc}{2}\right) d|\vec{p}_2| \qquad \Gamma = \frac{S}{16\pi\hbar m}|\mathcal{M}|^2$$

$\mathcal{M} \to M\left(\frac{m}{2}\right)$

1→ 2+3 Decay Massive m2=m3

$$\Gamma = \frac{S}{\hbar m}\left(\frac{1}{4\pi}\right)^2 \frac{1}{2}\int \frac{|\mathcal{M}|^2}{\underbrace{|\vec{p}_2||\vec{p}_3|}_{E_2 = E_3}}\,\delta\left(mc - \underbrace{|\vec{p}_2| - |\vec{p}_3|}_{E_{2,3}}\right)\delta^3(-\vec{p}_2 - \vec{p}_3)\, d^3\vec{p}_2\, \cancel{d^3\vec{p}_3}$$

$(\times 4\pi)$

$$\Gamma = \frac{S}{2(4\pi)^2 \hbar m}\int \frac{|\mathcal{M}|^2}{\underbrace{|\vec{p}_2|^2}_{E_2^2}}\,\delta\left(mc - 2|\vec{p}_2|\right) d^3\vec{p}_2 \;\rightarrow\; [\,''\,]\int \frac{\mathcal{M}^2}{E_2^2}\,\delta(mc - 2E_2)\, p_2^2\, dp_2$$

$$\frac{dE_2}{dp_2} = \frac{p_2}{E_2} \Rightarrow \int \mathcal{M}^2\,\delta(mc - 2E_2)\,\frac{p_2}{E_2}\, dE_2$$

$m_1/2$

$$\delta(kx) = \frac{1}{k}\delta(x)$$

$$\boxed{\Gamma = \frac{S|\mathbf{p}|}{8\pi\hbar m_1^2 c}|\mathcal{M}|^2}$$

p_2

$$p_2 = \sqrt{E_2^2 - m_2^2} = \sqrt{\left(\frac{m_1}{2}\right)^2 - m_2^2}$$

from last δ fn

Decay Rate Calculations

- For the two body decay

$$i \rightarrow 1 + 2$$

$$\begin{aligned} M_{fi} &= \langle \psi_1' \psi_2' | \hat{H}' | \psi_i' \rangle \\ &= (2E_i.2E_1.2E_2)^{1/2} \langle \psi_1 \psi_2 | \hat{H}' | \psi_i \rangle \\ &= (2E_i.2E_1.2E_2)^{1/2} T_{fi} \end{aligned}$$

★ Now expressing T_{fi} in terms of M_{fi} gives

$$\boxed{\Gamma_{fi} = \frac{(2\pi)^4}{2E_i} \int |M_{fi}|^2 \delta(E_i - E_1 - E_2) \delta^3(\vec{p}_a - \vec{p}_1 - \vec{p}_2) \frac{d^3\vec{p}_1}{(2\pi)^3 2E_1} \frac{d^3\vec{p}_2}{(2\pi)^3 2E_2}}$$

Note:

- M_{fi} uses relativistically normalised wave-functions. It is Lorentz Invariant
- $\dfrac{d^3\vec{p}}{(2\pi)^3 2E}$ is the Lorentz Invariant Phase Space for each final state particle
  the factor of $2E$ arises from the wave-function normalisation
  (prove this in Question 2)
- This form of Γ_{fi} is simply a rearrangement of the original equation
  but the integral is now frame independent (i.e. L.I.)
- Γ_{fi} is inversely proportional to E_i, the energy of the decaying particle. This is exactly what one would expect from time dilation ($E_i = \gamma m$).
- Energy and momentum conservation in the delta functions

Decay Rate Calculations

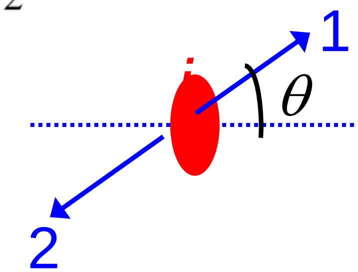
$$\Gamma_{fi} = \frac{(2\pi)^4}{2E_i}\int |M_{fi}|^2 \delta(E_i - E_1 - E_2)\delta^3(\vec{p}_i - \vec{p}_1 - \vec{p}_2)\frac{\mathrm{d}^3\vec{p}_1}{(2\pi)^3 2E_1}\frac{\mathrm{d}^3\vec{p}_2}{(2\pi)^3 2E_2}$$

★ Because the integral is Lorentz invariant (i.e. frame independent) it can be evaluated in any frame we choose. The C.o.M. frame is most convenient

- In the C.o.M. frame $E_i = m_i$ and $\vec{p}_i = 0$

$$\Gamma_{fi} = \frac{1}{8\pi^2 E_i}\int |M_{fi}|^2 \delta(m_i - E_1 - E_2)\delta^3(\vec{p}_1 + \vec{p}_2)\frac{\mathrm{d}^3\vec{p}_1}{2E_1}\frac{\mathrm{d}^3\vec{p}_2}{2E_2}$$

- Integrating over $\vec{p}_2$ using the δ-function:



$$\Gamma_{fi} = \frac{1}{8\pi^2 E_i}\int |M_{fi}|^2 \delta(m_i - E_1 - E_2)\frac{\mathrm{d}^3\vec{p}_1}{4E_1E_2}$$

now $E_2^2 = (m_2^2 + |\vec{p}_1|^2)$ since the δ-function imposes $\vec{p}_2 = -\vec{p}_1$

- Writing $\mathrm{d}^3\vec{p}_1 = p_1^2\mathrm{d}p_1 \sin\theta \mathrm{d}\theta \mathrm{d}\phi = p_1^2 \mathrm{d}p_1 \mathrm{d}\Omega$

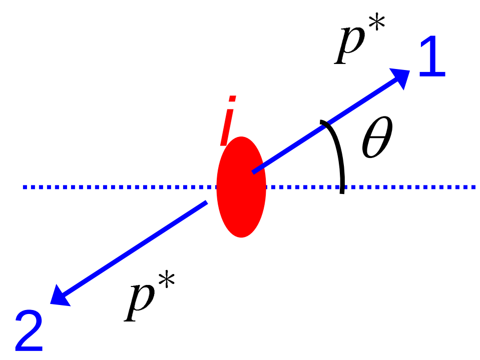
For convenience, here $|\vec{p}_1|$ is written as p_1

$$\Gamma_{fi} = \frac{1}{32\pi^2 E_i}\int |M_{fi}|^2 \delta\left(m_i - \sqrt{m_1^2 + p_1^2} - \sqrt{m_2^2 + p_1^2}\right)\frac{p_1^2 \mathrm{d}p_1 \mathrm{d}\Omega}{E_1E_2}$$

- Which can be written in the form $$\Gamma_{fi} = \frac{1}{32\pi^2 E_i}\int |M_{fi}|^2 g(p_1)\delta(f(p_1))\mathrm{d}p_1\mathrm{d}\Omega \qquad (2)$$

where $g(p_1) = p_1^2/(E_1E_2) = p_1^2(m_1^2+p_1^2)^{-1/2}(m_2^2+p_1^2)^{-1/2}$

and $f(p_1) = m_i - (m_1^2+p_1^2)^{1/2} - (m_2^2+p_1^2)^{1/2}$



Note:
- $\delta(f(p_1))$ imposes energy conservation.
- $f(p_1)=0$ determines the C.o.M momenta of the two decay products

  i.e. $f(p_1)=0$ for $p_1 = p^*$

★ **Eq. (2) can be integrated using the property of δ – function derived earlier (eq. (1))**

$$\int g(p_1)\delta(f(p_1))\mathrm{d}p_1 = \frac{1}{|\mathrm{d}f/\mathrm{d}p_1|_{p^*}}\int g(p_1)\delta(p_1-p^*)\mathrm{d}p_1 = \frac{g(p^*)}{|\mathrm{d}f/\mathrm{d}p_1|_{p^*}}$$

where p^* is the value for which $f(p^*)=0$

- All that remains is to evaluate $\mathrm{d}f/\mathrm{d}p_1$

$$\frac{\mathrm{d}f}{\mathrm{d}p_1} = -\frac{p_1}{(m_1^2+p_1^2)^{1/2}} - \frac{p_1}{(m_2^2+p_1^2)^{1/2}} = -\frac{p_1}{E_1} - \frac{p_1}{E_2} = -p_1\frac{E_1+E_2}{E_1E_2}$$

giving:

$$\Gamma_{fi} = \frac{1}{32\pi^2 E_i}\int |M_{fi}|^2 \left|\frac{E_1E_2}{p_1(E_1+E_2)}\frac{p_1^2}{E_1E_2}\right|_{p_1=p^*} \mathrm{d}\Omega$$

$$= \frac{1}{32\pi^2 E_i}\int |M_{fi}|^2 \left|\frac{p_1}{E_1+E_2}\right|_{p_1=p^*} \mathrm{d}\Omega$$

- But from $f(p_1)=0$, i.e. energy conservation: $E_1+E_2=m_i$

$$\Gamma_{fi} = \frac{|\vec{p}^*|}{32\pi^2 E_i m_i}\int |M_{fi}|^2 \mathrm{d}\Omega$$

In the particle's rest frame $E_i = m_i$

$$\left[\frac{1}{\tau} = \Gamma = \frac{|\vec{p}^*|}{32\pi^2 m_i^2}\int |M_{fi}|^2 \mathrm{d}\Omega\right] \quad (3)$$

VALID FOR ALL TWO-BODY DECAYS !

- p^* can be obtained from $f(p_1)=0$

$$(m_1^2+p^{*2})^{1/2}+(m_2^2+p^{*2})^{1/2}=m_i \quad \text{(Question 3)}$$

$$p^* = \frac{1}{2m_i}\sqrt{\left[(m_i^2-(m_1+m_2)^2\right]\left[m_i^2-(m_1-m_2)^2\right]} \quad \text{(now try Questions 4 \& 5)}$$

Simplified versions

- Cross section 1+2→3+4 → SPINLESS

$$\frac{d\sigma}{d\Omega} = \left(\frac{\hbar c}{8\pi}\right)^2 \frac{S|\mathcal{M}|^2}{(E_1 + E_2)^2} \frac{|\mathbf{p}_f|}{|\mathbf{p}_i|}$$

s

Better invariant version

$$\boxed{\frac{\mathrm{d}\sigma}{\mathrm{d}t} = \frac{1}{64\pi s|\vec{p}_i^{\,*}|^2}|M_{fi}|^2}$$

- THE ANGULAR DISTRIBUTION IS SOLELY DETERMINED BY THE Matrix Element !!!

Simplifcations for Cross Section/ Similar

- Probably won't go through similar simplification in Griffiths for 1+2→3+4 Scattering (?)

For the process 1 + 2 → 3 + 4 + …. + N

$$d\sigma = |\mathcal{M}|^2 \frac{\hbar^2 S}{4\sqrt{(p_1 \cdot p_2)^2 - (m_1 m_2 c^2)^2}} \left[\left(\frac{c\, d^3\vec{p}_3}{(2\pi)^3 2E_3}\right)\left(\frac{c\, d^3\vec{p}_4}{(2\pi)^3 2E_4}\right) \cdots\cdots \left(\frac{c\, d^3\vec{p}_n}{(2\pi)^3 2E_n}\right)\right] \times (2\pi)^4 \delta^4(p_1 + p_2 - \ldots - p_n)$$

$$\frac{d\sigma}{d\Omega} = \left(\frac{\hbar c}{8\pi}\right)^2 \frac{S|\mathcal{M}|^2}{(E_1 + E_2)^2} \frac{|\mathbf{p}_f|}{|\mathbf{p}_i|}$$

- But few comments about that process

$$\times \prod_{j=3}^{n} 2\pi\,\delta\left(p_j^2 - m_j^2 c^2\right)\theta\left(p_j^0\right)\frac{d^4 p_j}{(2\pi)^4}$$

whole big thing just $\equiv d^3p/E$

- First as said before use the above not Griffs for the master formula
- instead of d^3p/E , his uses more obsfuscated/way less common d^4p*stuff
- One very important part of the simplf. process used above explained in Griff:
- Now there are more integrals, less delta functions – how do we again are we able to avoid integrating over the M^2 ?
  - Answer: Instead of doing the angular integrals, move the angular part (or more generally can just move one of the whole d^3p/E for invariant x.s.)

$$d\sigma = |\mathcal{M}|^2 \frac{\hbar^2 S}{4\sqrt{(p_1 \cdot p_2)^2 - (m_1 m_2 c^2)^2}} \left[\boxed{\left(\frac{c\, d^3\vec{p}_3}{(2\pi)^3 2E_3}\right)}\right]$$

σ

$p_3^2\, dp_3\, d\Omega$

$\frac{d\sigma}{d\Omega}$

$E \frac{d\sigma}{d^3 p_3}$

$\left(\frac{d\sigma}{p_T\, dp_T\, dy}\right)$

could be 4 (or 5, 6, 7...)

Flux

About the flux factor– I may have misspoken before. It has more to it than simply Lorentz Invariance although that does play a role obviously—apparent from its form.

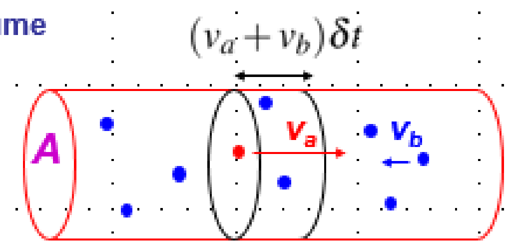
For the process 1 + 2 ➔ 3 + 4 + …. + N

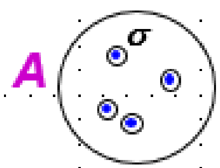
$$d\sigma = |\mathcal{M}|^2 \frac{\hbar^2 S}{4\sqrt{(p_1\cdot p_2)^2-(m_1 m_2 c^2)^2}}\left[\left(\frac{c\,d^3\vec{p}_3}{(2\pi)^3 2E_3}\right)\left(\frac{c\,d^3\vec{p}_4}{(2\pi)^3 2E_4}\right)\cdots\cdots\left(\frac{c\,d^3\vec{p}_n}{(2\pi)^3 2E_n}\right)\right]\times(2\pi)^4\delta^4(p_1+p_2-\ldots\ldots-p_n)$$

The reason it is literally from "beam" flux – see here::

- Consider a single particle of type ***a*** with velocity, $\boldsymbol{V_a}$, traversing a region of area ***A*** containing $\boldsymbol{n_b}$ particles of type ***b*** per unit volume

In time δt a particle of type a traverses region containing $n_b(v_a+v_b)A\delta t$ particles of type ***b***





★Interaction probability obtained from effective cross-sectional area occupied by the $n_b(v_a+v_b)A\delta t$ particles of type ***b***

- Interaction Probability = $\dfrac{n_b(v_a+v_b)A\delta t\sigma}{A} = n_b v\delta t\sigma$ $\quad [v = v_a+v_b]$

➡ $\boxed{\text{Rate per particle of type } a = n_b \, v\, \sigma}$

- Consider volume ***V***, total reaction rate = $(n_b v\sigma).(n_a V) = (n_b V)(n_a v)\sigma$
$= N_b\phi_a\sigma$

we shall see that flux factor is essentially this $\sim n_a n_b(va+vb)$ (n means N/V)

Interesting point: va+vb does not need relativistic addition here – it can be 2c ! b.c. the rate of crossings in our frame is not limited by the S.R. constraint, va+vb even for both $v_i \sim c$

Flux

The scattering Golden Rule has this non-trivial looking factor we call the flux – why is to be discussed next time

For the process 1 + 2 ➔ 3 + 4 + …. + N

$$d\sigma = \left|\mathcal{M}\right|^2 \frac{\hbar^2 S}{4\sqrt{(p_1 \cdot p_2)^2 - (m_1 m_2 c^2)^2}} \left[\left(\frac{c d^3\vec{p}_3}{(2\pi)^3 2E_3}\right)\left(\frac{c d^3\vec{p}_4}{(2\pi)^3 2E_4}\right) \cdots\cdots \left(\frac{c d^3\vec{p}_n}{(2\pi)^3 2E_n}\right)\right] \times (2\pi)^4 \delta^4(p_1 + p_2 - \ldots\ldots - p_n)$$

From Thomson's (Roche) slides we see a nice couple cases of this factor written in different perhaps more recognizable forms. These are useful to use along with the formula above itself.

$$\sigma = \frac{(2\pi)^{-2}}{2E_1 2E_2 (v_1 + v_2)} \int |M_{fi}|^2 \delta(E_1 + E_2 - E_3 - E_4) \delta^3(\vec{p}_1 + \vec{p}_2 - \vec{p}_3 - \vec{p}_4) \frac{d^3\vec{p}_3}{2E_3} \frac{d^3\vec{p}_4}{2E_4}$$

- The integral is now written in a Lorentz invariant form
- The quantity $F = 2E_1 2E_2 (v_1 + v_1)$ can be written in terms of a four-vector scalar product and is therefore also Lorentz Invariant (the Lorentz Inv. Flux)

$$F = 4\left[(p_1^{\mu} p_{2\mu})^2 - m_1^2 m_2^2\right]^{1/2}$$

(see appendix I)

- Consequently cross section is a Lorentz Invariant quantity

Two special cases of Lorentz Invariant Flux:

- Centre-of-Mass Frame

$$\begin{aligned} F &= 4E_1E_2(v_1 + v_2) \\ &= 4E_1E_2(|\vec{p}^*|/E_1 + |\vec{p}^*|/E_2) \\ &= 4|\vec{p}^*|(E_1 + E_2) \\ &= 4|\vec{p}^*|\sqrt{s} \end{aligned}$$

- Target (particle 2) at rest

$$\begin{aligned} F &= 4E_1E_2(v_1 + v_2) \\ &= 4E_1 m_2 v_1 \\ &= 4E_1 m_2 (|\vec{p}_1|/E_1) \\ &= 4m_2|\vec{p}_1| \end{aligned}$$

Example Toy (ABC) Theory Feynman Rules

- Feunman Rules to Calculate M

Our problem is to find the amplitude $\mathcal{M}$ associated with a given Feynman diagram. The ritual is as follows [5]:

1. *Notation*: Label the incoming and outgoing four-momenta $p_1, p_2, \ldots, p_n$ (Figure 6.6). Label the internal momenta $q_1, q_2, \ldots$. Put an arrow beside each line, to keep track of the 'positive' direction (forward in time for external lines, arbitrary for internal lines).
2. *Vertex factors*: For each vertex, write down a factor

$$-ig$$

g is called the *coupling constant*; it specifies the strength of the interaction between A, B, and C. In this toy theory, g has the dimensions of momentum; in the 'real-world' theories, we shall encounter later on, the coupling constant is always dimensionless.

3. *Propagators*: For each internal line, write a factor

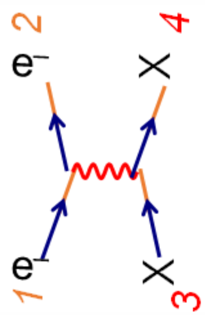
$$\frac{i}{q_j^2 - m_j^2 c^2}$$

where q_j is the four-momentum of the line and m_j is the mass of the particle the line describes. (Note that $q_j^2 \neq m_j^2 c^2$, because a virtual particle does not lie on its mass shell.)

4. *Conservation of energy and momentum*: For each vertex, write a delta function of the form

$$(2\pi)^4 \delta^4(k_1 + k_2 + k_3)$$

where the k's are the three four-momenta coming *into* the vertex (if the arrow leads outward, then k is *minus* the four-momentum of that line). This factor imposes conservation of energy and momentum at each vertex since the delta function is zero unless the sum of the incoming momenta equals the sum of the outgoing momenta.



5. *Integration over internal momenta*: For each internal line, write down a factor*

$$\frac{1}{(2\pi)^4} d^4 q_j$$

and integrate over all internal momenta.

6. *Cancel the delta function*: The result will include a delta function

$$(2\pi)^4 \delta^4(p_1 + p_2 + \cdots - p_n)$$

reflecting overall conservation of energy and momentum. Erase this factor† and multiply by i. The result is $\mathcal{M}$.

<u>Summarized</u>

- 1) Diagram – Notate (1,2,3,.etc)
- 2) Each Vertex factor F – coupling g → F = -ig
- 3) (Each) propagator $i/(q^2-m^2)$
- 4) Eac vertex ("uvw") $\delta(p_u \pm p_v \pm p_w)$ ie 4v mom conservation 3legs/arrows
- 5) Integrate internal lines d^4q
- 6) Cancel/erase the final delta fn (δ^4) – global 4v mom consv
- 7) Multiply by i

tree diagram δfn's: 2 (8) δfns [two 4v]- 1(4) Integrals-1 (4) final = no "real" integrations!